Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. All experiments were run on a single GPU with 16 GB of memory.

2 Results

Figure 1 shows the f1 score of LSTM by dataset.

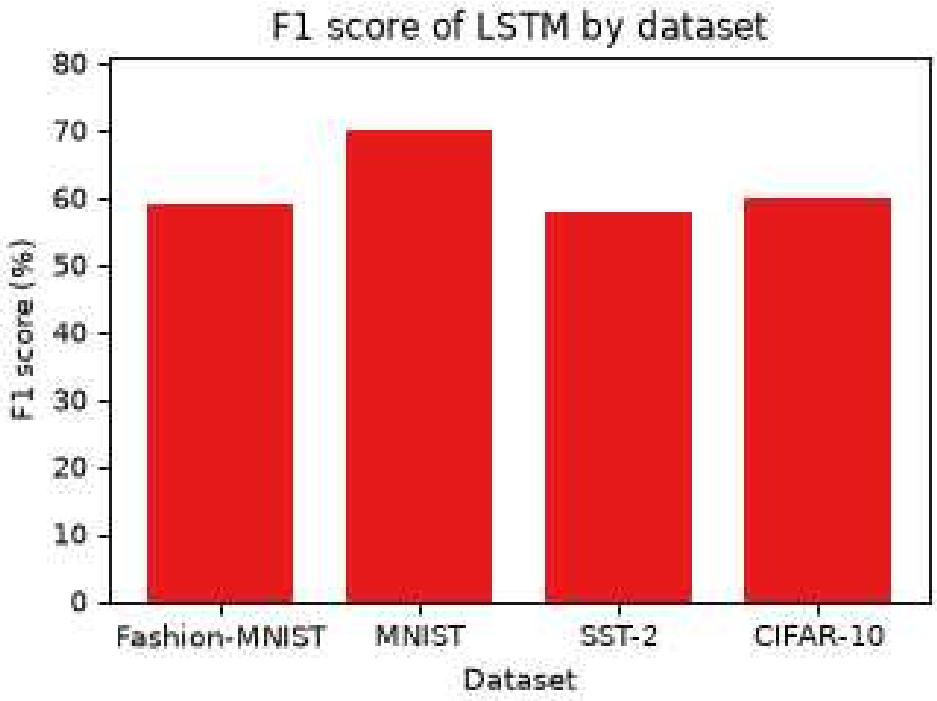


Figure 1: F1 score of LSTM by dataset for the models in this experiment.

LSTM scores 16.4 points higher on MNIST than on SST-2. As shown in Figure 1, LSTM reaches 59.1% F1 score on Fashion-MNIST.

Figure 2 shows the test accuracy of MLP by dataset.

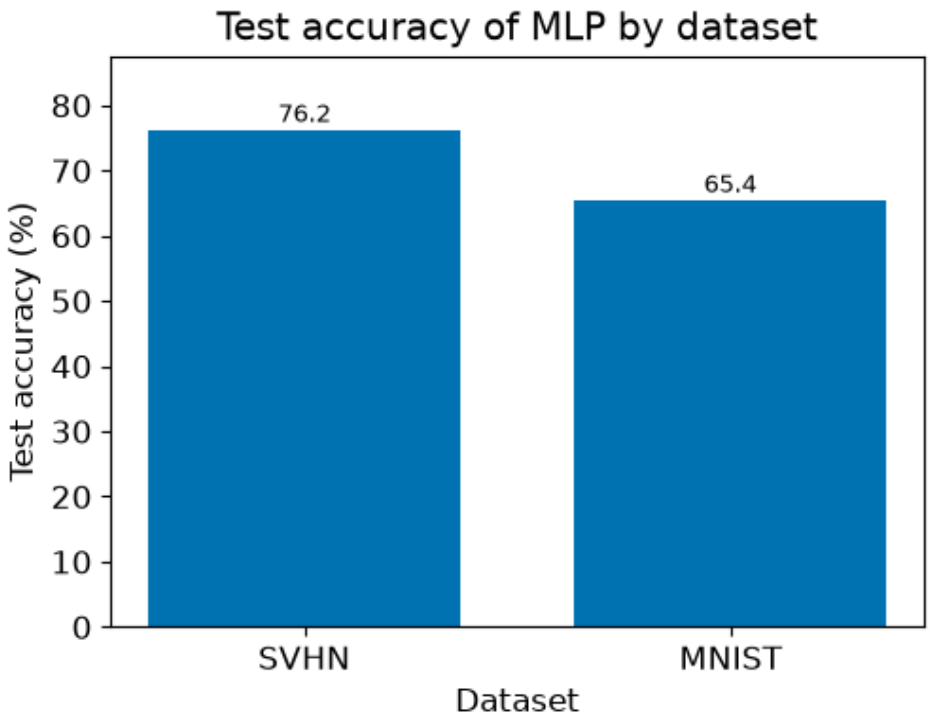


Figure 2: Test accuracy of MLP by dataset for the models in this experiment.

On MNIST, MLP obtains a test accuracy of 109.2% (Figure 2). MLP reaches 65.4% test accuracy on MNIST. MLP achieves 32.4% test accuracy on SVHN. On MNIST, MLP obtains a test accuracy of 82.9%.

Figure 3 shows the validation accuracy of ViT-S by dataset.

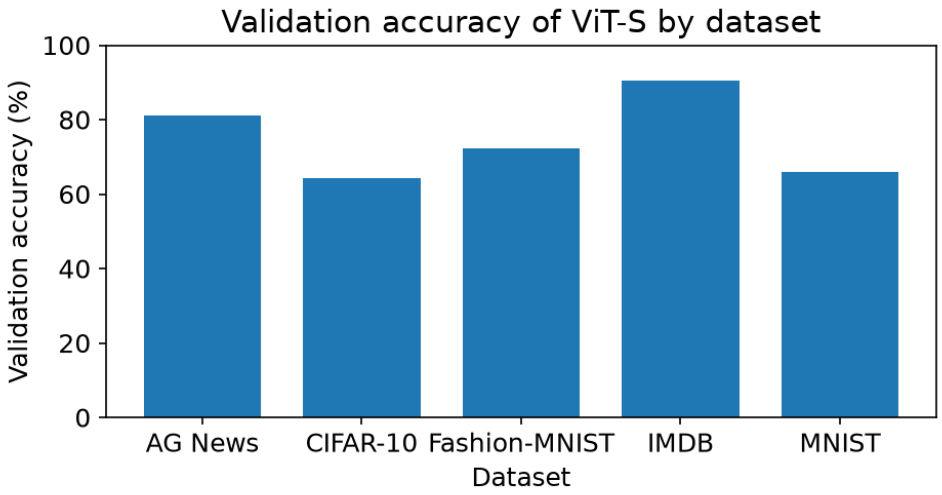


Figure 3: Validation accuracy of ViT-S by dataset for the models in this experiment.

ViT-S scores 58.8 points higher on AG News than on Fashion-MNIST. On IMDB, ViT-S obtains a validation accuracy of 72.2%. ViT-S reaches 90.4% validation accuracy on IMDB.

3 Discussion

Random Forest reaches 91.4% accuracy on Fashion-MNIST. Overall, the results are in line with what we expected from the literature.